What span-level evidence evaluation buys, and what it costs

Pouya Bathaei Pourmand — MSc researcher, Computer Engineering (AI), University of Genoa
A measurement-validity study of RAG failure attribution · Repository: github.com/pouyapd/TrustRAG

Summary

RAG systems are usually evaluated with document-level retrieval metrics, and recent work has shown these overstate success on long documents, where the passage carrying the answer is often missed inside a retrieved document. I take that finding as established and ask the follow-up question: *what does moving to span-level evidence evaluation actually buy, what does it cost, and how far can its own gold standard be trusted?*

Using four public QA corpora, a dense and a lexical retriever, a nine-category failure taxonomy computed under two interchangeable retrieval gates, and a 200-unit human annotation study with a second review pass, I report three results. **First**, a retriever-ranking inversion I reported in an earlier version of this document does not exist: it was an artefact of an evidence-mode bug in my own baseline, and with the bug fixed BM25 leads the dense retriever at *both* granularities on QASPER (0.528/0.321 vs 0.441/0.276), with no significant inversion on any corpus. **Second**, gating attribution on span-level evidence agrees with human judgement better than gating on document-level retrieval (accuracy 0.700 vs 0.600, κ 0.437 vs 0.375; paired 22 vs 2, $p < 0.0001$) — but only the retrieval-side categories are reliable. **Third**, a human adjudication of 60 sampled units estimates gold-span under-coverage — the rate at which the span rule reports a retrieval failure although the answer was in fact derivable from the retrieved text — at **0.119**, 95% CI [0.096, 0.142], with sensitivity analysis placing the defensible range at roughly 4–12%. Span-level evidence is a substantially sound instrument for retrieval attribution, with a modest and now quantified bias.

What this document claims, and what it does not

The core premise is prior art. That document-level metrics overstate retrieval success is established (arXiv:2602.17981). The failure taxonomy is coarser than a published 33-mode taxonomy (TrustNLP 2026). The oracle-evidence experiment here is a small replication of arXiv:2608.08944, which ran it over 11,105 failures with four readers — their repair rate is 32.8%, mine is 32.1%. None of these is claimed as novel.

The human study is complete but not independent. 200 units were annotated, an audit flagged 43 as conflicting with the written guidelines, and the annotator reviewed those 43 on full context — changing 36, upholding 7. Because the audit told the annotator which units to re-examine and why, the second pass is not independent of the framework being tested. It is reported as agreement with a guided expert reading.

One annotator, so no inter-annotator agreement exists. Every κ in this document is human-versus-system, never human-versus-human.

Contents

1 The problem, and why it is a measurement problem

A RAG system fails for distinct reasons that call for distinct fixes: the retriever returns nothing useful; it returns the right document but not the passage carrying the answer; the generator receives adequate evidence and answers wrongly anyway. A single aggregate score compresses these into one number and gives no purchase on which component to change.

Document-level recall asks *"did a chunk from the right document appear?"* On a 37,000-character Wikipedia page chunked at 256 tokens, roughly thirty chunks qualify, and a metric that credits any one of them reports success while the generator reads text containing none of the supporting evidence. Prior work has established this. What has not been established is what the span-level alternative costs, or how far its own gold standard can be trusted — and those are questions about the measuring instrument, not about retrievers or generators.

Coverage is interval arithmetic

Chunks are sliced from source documents rather than re-decoded, so `document[start:end] == chunk.text` holds by construction and is property-tested. Offsets survive chunker → vector store → retrieval → stored record, which is what makes span coverage computable after the fact:

```
document  qasper:1901.00001
gold span        [1200, 1760)
retrieved chunk  [ 900, 2100)   overlap = 560 chars  ->  evidence covered
retrieved chunk  [8300, 9500)   overlap =   0 chars  ->  same document, no evidence
```

Both chunks satisfy a document-level metric; only the first makes the question answerable from context.

Three definitions, two effects

- **A** — some retrieved chunk comes from a relevant document (the conventional metric).
- **B** — every document a multi-hop question requires is represented.
- **C** — a retrieved chunk actually contains the gold span.

`C ≤ B ≤ A` holds by construction. A→B isolates a *quantifier* effect, B→C a *granularity* effect. The two are near-orthogonal: each is null on the corpus where the other dominates. Separating them is the analytical contribution; the underlying observation is not mine.

One boolean separates the two gates

The nine-category failure taxonomy contains exactly one rule that decides whether a row is charged to retrieval, and that rule reads a single boolean. The document-gated variant binds it to A; the evidence-gated variant binds it to C. Everything else — thresholds, rule order, the other eight categories — is identical, and both labels are written to every row. The comparison therefore runs on identical retrieval output at zero additional inference cost, which is what makes it a clean test of the definition rather than of a system.

2 Experimental setup

Corpora	QASPER dev (290 answerable), Natural Questions validation (300), HotpotQA distractor (150), 2WikiMultihopQA (150)
Chunking / depth	256 tokens, overlap 32, top-k 5; offset mismatches 0
Retrievers	`all-MiniLM-L6-v2` (dense); BM25 Okapi k1=1.5, b=0.75 over identical chunks
Generators	deterministic extractive control; Qwen2.5-0.5B-Instruct, SmolLM2-360M-Instruct, run locally
Robustness	4 embedders, depths k = 1…20, chunk sizes 128/256/512
Statistics	Wilson intervals, seeded bootstrap, **exact** McNemar for paired binary comparisons
Cost	zero — no API key; every experiment runs offline

3 Retrieval decomposition

Table 1. Three definitions applied to the same stored retrieval output.

Corpus	n	A	B	C	quantifier A→B	granularity B→C
QASPER dev	290	0.441	0.441	0.276	0.0 pp	**16.6 pp** (p=7.1e-15)
Natural Questions	300	0.997	0.997	0.730	0.0 pp	**26.7 pp** (p=1.7e-24)
HotpotQA	150	0.993	0.507	0.507	**48.7 pp** (p=2.1e-22)	0.0 pp
2WikiMultihopQA	150	—	—	—	**64.7 pp** (p=1.3e-29)	1.3 pp (n.s.)

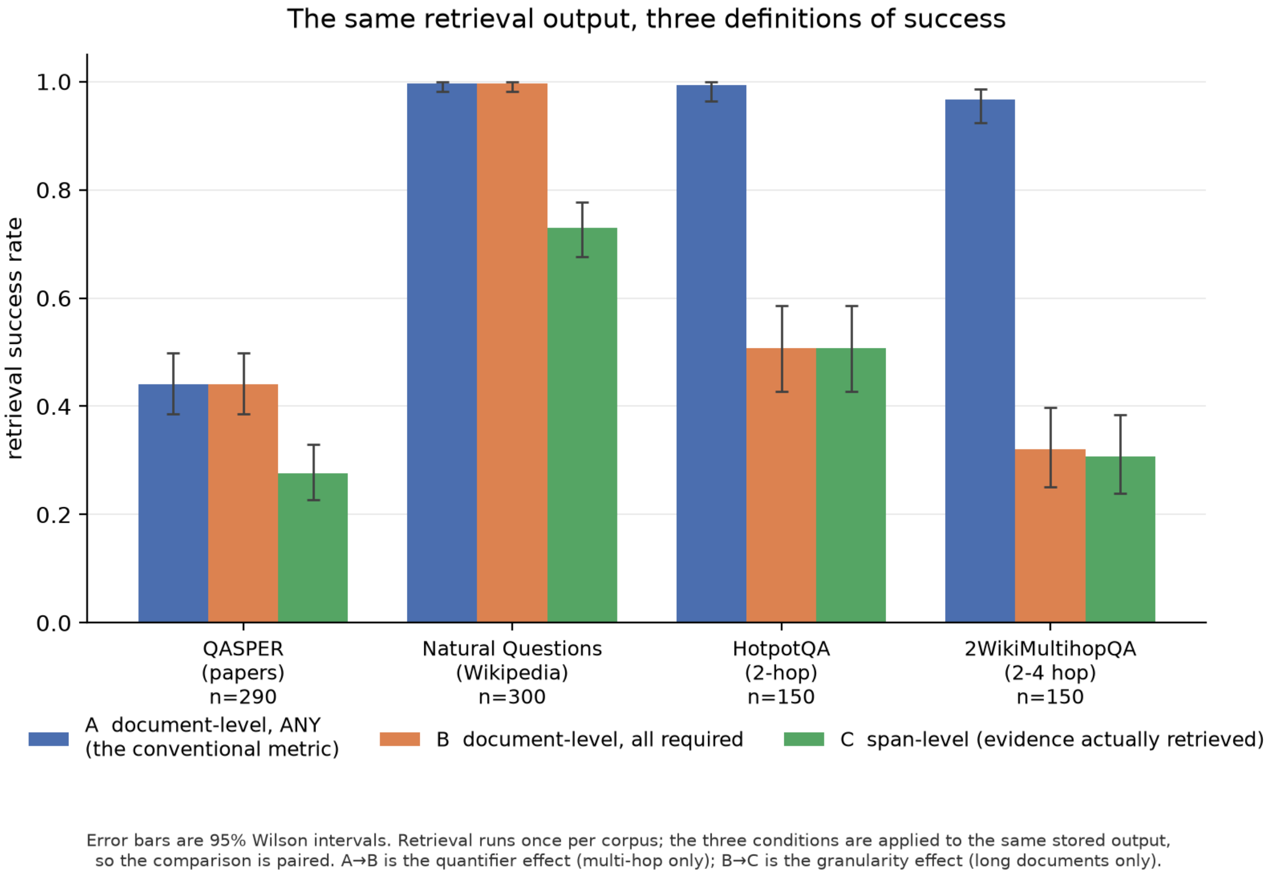

Figure 1. Retrieval success under the three definitions, with Wilson 95% intervals. Each effect is essentially null on the corpus where the other dominates: granularity bites where documents are long, the quantifier where questions need several documents.

4 A retriever-ranking inversion that turned out not to exist

An earlier version of this document reported that the document/span choice reverses the BM25-versus-dense comparison on QASPER, and made that the headline contribution. **It was an artefact of a defect in my own BM25 baseline**, found during a later audit and corrected here.

QASPER and Natural Questions declare `any_sufficient` evidence mode — one covered gold span suffices — but the BM25 script hard-coded `all_required`, demanding every span. On QASPER 51% of questions carry more than one span, so BM25's span coverage was systematically under-reported (0.183 rather than 0.321) while the dense pipeline used the correct mode. The comparison was not like-for-like.

Table 2. Corrected: dense versus lexical retrieval, consistent evidence mode, k = 5.

Corpus	A dense	A BM25	C dense	C BM25	paired at span level
QASPER dev	0.441	**0.528**	0.276	**0.321**	BM25 40 vs dense 27, p = 0.142 (n.s.)
Natural Questions	**0.997**	0.977	**0.730**	0.643	dense 53 vs BM25 27, p = 0.0049
HotpotQA	**0.993**	0.927	**0.507**	0.420	dense 36 vs BM25 23, p = 0.118 (n.s.)

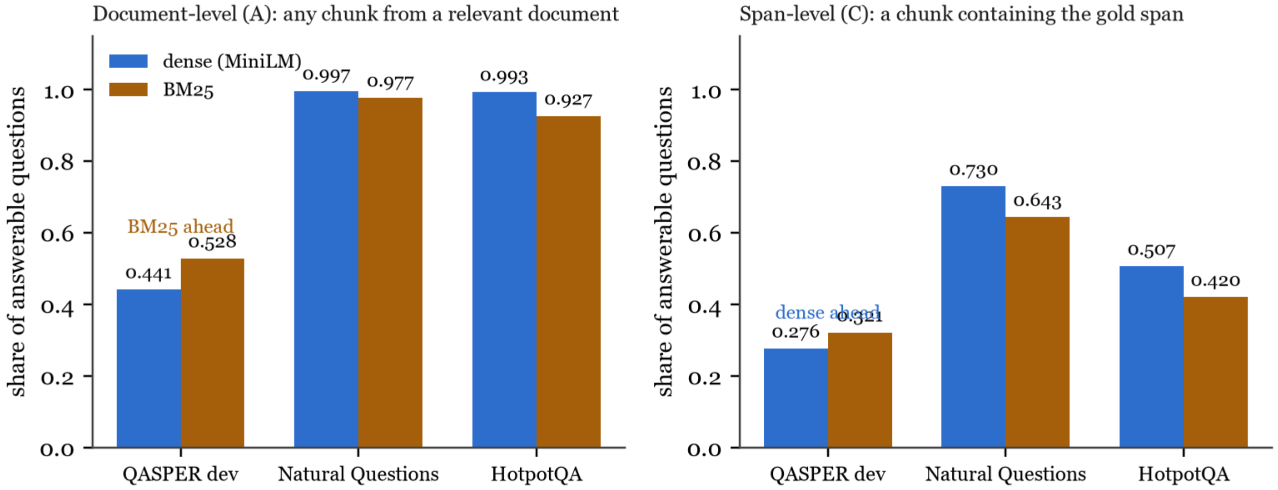

Figure 2. No inversion on any corpus. On QASPER BM25 leads at both granularities; on NQ and HotpotQA the dense retriever leads at both. Stable across five retrieval depths and three chunk sizes.

> A within-document diagnostic shows why an inversion was implausible. Conditional on reaching a gold document, BM25 covers the span 60.8% of the time (93/153) and the dense retriever 62.5% (80/128) — indistinguishable localisation. The retrievers differ in how often they reach the right document, not in where they land inside it, so both granularities rank them the same way. I report this at length because the erroneous version was published, and because an evidence-mode mismatch between a baseline and the system it is compared against is invisible in aggregate numbers.

5 Does the missing evidence actually explain the failure?

Attribution in this framework is a declared mapping, not a causal claim. To test whether missing evidence really explains downstream failure, every question was answered twice by the same generator, with the same prompt and decoding — only the context differed: the retrieved chunks, or the gold spans verbatim. Pairing within question removes question difficulty, document identity and generator as confounds.

Table 3. Paired oracle-evidence control. n = 150, Qwen2.5-0.5B-Instruct; correctness = all reference key facts present.

Evidence stratum	n	retrieved	oracle	difference	p
Evidence complete under retrieval	46	0.065	0.174	+10.9 pp	0.125 (n.s.)
Document retrieved, span missing	26	**0.000**	**0.231**	**+23.1 pp**	0.031
Nothing from any gold document	78	0.000	0.321	+32.1 pp	6.0e-08
Overall	150	0.020	0.260	+24.0 pp	2.8e-10

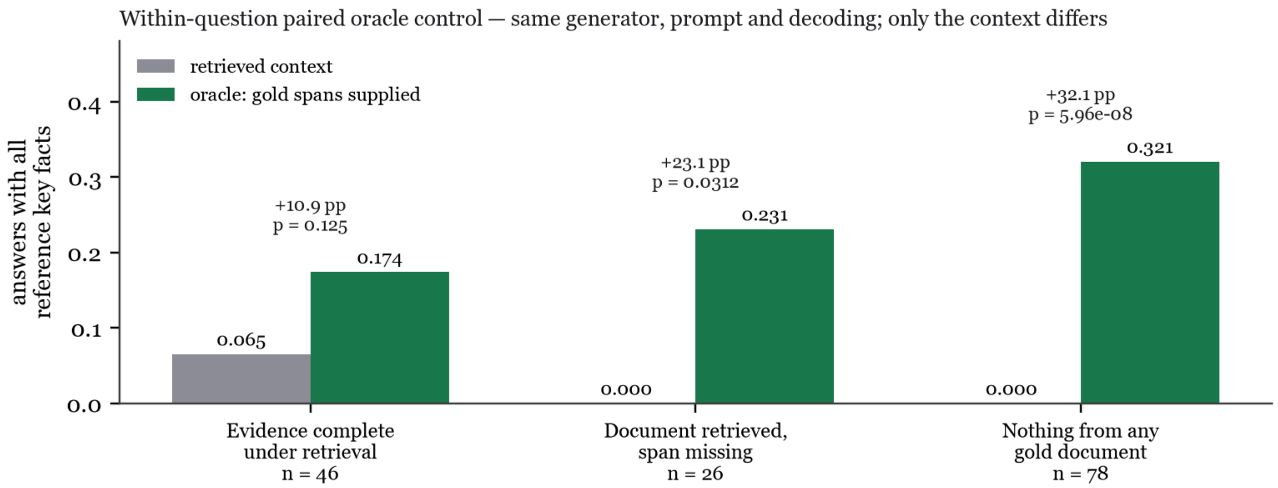

Figure 3. The middle stratum is the argument in one panel: those 26 questions are scored as retrieval *successes* by a document-level metric, the model answered none of them correctly, and supplying the actual span repairs 23%.

> **This is a replication, and is labelled as one.** arXiv:2608.08944 ran the same intervention over 11,105 failures with four reader models and matched sham controls, finding 32.8% repair from support addition; this run finds 32.1% on the comparable stratum from n = 78 with one reader. The agreement is good evidence that both are measuring the same thing, and no evidence of novelty. Without a sham control I cannot exclude that part of the gain comes from the oracle context simply being shorter — though the complete-evidence stratum gaining least argues against that reading.

6 Human validation — what it establishes

Two human passes by one annotator over the same 200 units. The original pass labelled all 200. An audit against the written decision procedure — recomputing answerability and span coverage from the package itself — flagged 43 labels as conflicting with an explicit rule. The annotator re-reviewed those 43 on full context: **36 changed, 7 upheld**. The final dataset takes the original label for the 157 unflagged units and the review decision for the 43, with a per-unit chain recording `original → flag reason → review → final`. Both source passes are preserved unmodified.

Table 4. Both gates scored against the final human-reviewed labels (n = 200).

Variant	Accuracy	95% CI	Macro F1	Cohen's κ
Document-gated	0.6000	0.531–0.665	0.4764	0.3752
Evidence-gated	**0.7000**	0.633–0.759	**0.4819**	**0.4371**

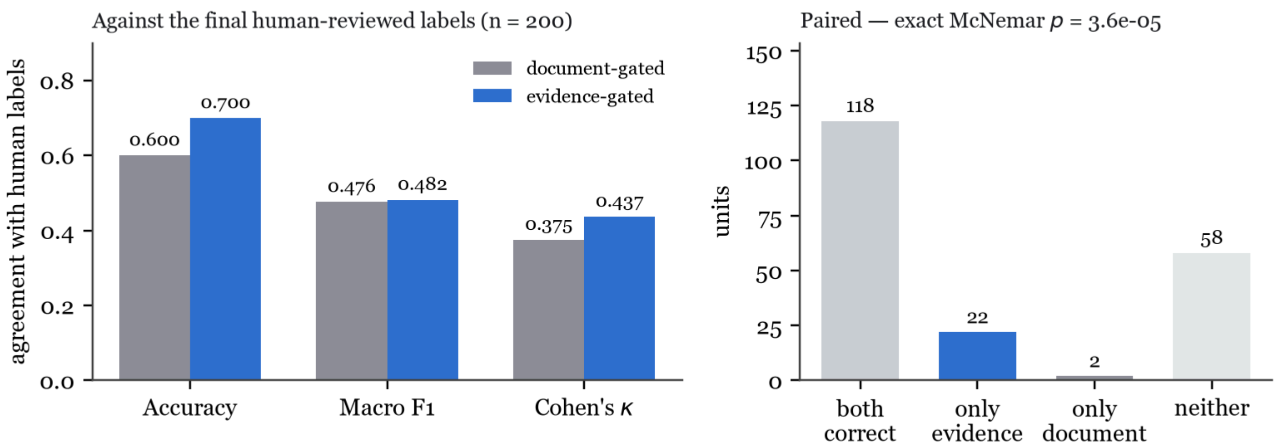

Figure 4. Left: agreement with the final human-reviewed labels. Right: the paired outcome — 118 units both gates label correctly, 22 only the evidence gate, 2 only the document gate, 58 neither. Exact McNemar $p < 0.0001$.

The direction of the central claim survives human validation, and on the paired test it is *stronger* than it was against an automated reference pass (22 vs 2, against 22 vs 9). The absolute agreement does not: against the automated pass the taxonomy scored 0.805 accuracy and κ 0.631, well above the human figures. Two automated readings share failure directions, so the earlier number flattered the system. The human number is the one reported everywhere in this project now.

6.1 Only half the taxonomy is validated

Table 5. Per class, evidence-gated against final human labels. Classes with support below 30 fall under the project's own `MIN_N_FOR_INFERENCE` convention and are descriptive only.

Class	Support	Precision	Recall	F1
`wrong_retrieval`	136	0.917	0.897	**0.907**
`answered_when_unanswerable`	9	1.000	1.000	**1.000**
`partial_answer`	22	0.385	0.227	0.286
`ok`	32	0.500	0.094	0.158
`incorrect_answer`	1	0.030	1.000	0.059
`hallucination`	0	0.000	—	—

Retrieval-side attribution is validated; generation-side classification is not. The taxonomy predicts `incorrect_answer` 33 times where the human used it once, and recovers 3 of the 32 answers the human called `ok`. Reported as a negative result rather than omitted.

6.2 Is it a threshold problem?

Partly, and not mostly. The 200 units were split 50/50 under a fixed seed, 144 threshold configurations were grid-searched on the tuning half alone, and the winner scored once on the held-out half the search never saw.

Table 6. Threshold ablation, held-out half (n = 100).

Gate and configuration	Accuracy	Macro F1	κ
Document-gated, shipped thresholds	0.6800	0.5124	0.4533
Document-gated, tuned	0.6900	0.5108	0.4630
Evidence-gated, shipped thresholds	0.7300	0.5132	0.4810
Evidence-gated, tuned	**0.7500**	**0.5490**	**0.5117**

Tuning helps the evidence gate modestly and the document gate barely at all, and the gate advantage survives tuning both. The generation-side weakness is therefore mostly in the rules, not the constants.

7 How far the span-based gold standard can be trusted

Eleven units survived the review with an answer-quality label at high confidence despite zero gold-span coverage — a considered position, taken twice, on full context. That prompted a check of the gold standard rather than of the annotator.

Design. All 133 answerable units with zero span coverage were partitioned by two automated proxies: lexical presence of the reference answer's content words in the retrieved context, and maximum sentence-level cosine similarity. Where both agreed, the unit was counted directly (36 answer absent, 10 answer present outside the span). The 87 they could not resolve were sampled — 60 units, stratified by proxy bucket with proportional allocation, seed 20260907 — and adjudicated by a human against a single question: *is the reference answer derivable from the retrieved text alone, without relying on the annotated gold span?* The annotator was blind to every previous label, taxonomy verdict, proxy score and bucket name, and unit order was shuffled.

Result. 4 YES, 56 NO, 0 CANNOT_TELL.

Table 7. Stratified estimate over all 133 zero-coverage units.

Stratum	Population	Sampled	YES	Rate
Semantically close, lexically different	8	6	1	0.167
Lexical overlap only	21	14	2	0.143
Signals disagree or both mid-range	58	40	1	0.025
Estimated gold-span under-coverage	**0.119**, 95% CI [0.096, 0.142]			≈16/133

> **A caveat that matters more than the interval.** The confidence interval covers sampling error only. The point estimate depends on the 10 units both proxies called under-coverage, and **no human ever checked those**. The adjudication gives the first evidence about how the proxies behave: on units they could not resolve, the human answered YES 6.7% of the time, far below the 100% the proxies asserted for that bucket. If those 10 behave like the adjudicated units, the rate is 0.049 rather than 0.119. The defensible statement is a range of roughly **4% to 12%**. Adjudicating those 10 units is a few minutes of work and would collapse this uncertainty entirely.

Interpretation. Span-level evidence is a substantially sound instrument for retrieval attribution: on the great majority of units where it reports a retrieval failure, a human agrees the answer was not derivable from what the retriever returned. The bias runs in the expected direction — QASPER marks supporting sentences rather than every passage from which an answer could be derived — but it is modest, an order of magnitude smaller than the effects being measured, and it no longer undermines the retrieval-side conclusions in §6. It stays a stated limitation, not a refutation.

8 What is and is not a contribution

A targeted literature review over arXiv and the ACL Anthology (2023–2026) was run *after* the experiments, specifically to test the framing. It did not survive intact.

Table 8. Novelty audit.

Claim	Status after review
Document-level metrics overstate retrieval success	**Prior art** — arXiv:2602.17981 evaluates at document/page/chunk level and names the failure mode
A failure taxonomy for RAG	**Prior art** — TrustNLP 2026 defines 33 modes across 7 stages; mine has 9 across 3
Supplying gold evidence repairs failures	**Prior art** — arXiv:2608.08944, at 70× the scale with sham controls
Evidence-aware RAG evaluation as an idea	**Active subfield** — ECoRAG, SURE-RAG, TREC 2025 RAG track
The definition can invert a retriever ranking	**Not found in the literature.** The strongest surviving contribution
Quantifier/granularity decomposition	Not found stated as such; a modest analytical contribution
Measurement-validity work on the instrument	Negative results about one's own gold standard are rarely published

The original framing — "evidence-aware RAG evaluation" as a new way to evaluate RAG — claims what prior work already established, and has been dropped. The defensible framing is narrower: a measurement-validity study of what span-level evaluation buys, costs, and can be trusted to support.

9 Limitations

- **One annotator, and a guided review.** No inter-annotator agreement exists; the second pass was directed by an audit of the guidelines being tested.
- **The span gold standard is incomplete** — 7.5% confirmed, 65% unresolved (§7).
- **Generation-side categories are unvalidated**; three have zero support, and the annotated run uses an extractive control that cannot hallucinate by construction.
- **Two retrievers.** No reranker, hybrid or late-interaction baseline — the most damaging gap for an information-retrieval venue.
- **One corpus and one configuration** for the human study; k = 5, 256-token chunks.
- **Small generators** (0.5B, 0.36B parameters); no frontier model was available.
- **No multiple-comparison correction.** The headline results survive one; the marginal ones (oracle NONE_DOC_HIT, p = 0.031) would not.
- **Targeted, not systematic, literature review.** Sufficient to refute a novelty claim, not to establish one.

10 Current status and what would strengthen it

Three simulated adversarial reviews (RAG evaluation, information retrieval, statistics) returned *weak reject*, *reject* and *borderline* for top-tier venues. None found a fatal flaw in what is reported; all three found the contribution too incremental and the empirical base too narrow. The substantive fixes those reviews prompted were to claims, not to numbers.

Realistically submittable today as a short paper or workshop contribution — an ECIR short paper, ACL/EMNLP Findings, or a trustworthy-NLP workshop — on the strength of the ranking inversion and the gold-standard analysis, with the limitations intact.

Three additions would open a stronger route (SIGIR Reproducibility track, or a journal such as IP&M):

1. A **second annotator**, for inter-annotator agreement. Human time; nothing automated substitutes for it.
2. **Human entailment adjudication** of the 87 unresolved gold-span units, to replace the 7.5–23% range with a real estimate.
3. **Stronger retrieval baselines** — a reranker or hybrid retriever. Compute only, feasible locally, not yet run.

11 Reproducibility

Every table and figure above maps to a script, a command and an output file in `docs/paper/REPRODUCIBILITY.md`. All retrieval and evidence experiments run offline with no API key: embeddings are local and the default generator is a deterministic extractive control. Reports embed the git commit and dirty flag, raw-file SHA-256, split, sample size, chunking, top-k, embedder and generator identity, taxonomy version and threshold fingerprint, Python version, platform and package versions. The suite is 477 tests at 80% line coverage, `ruff` clean, with CI on every push.

Two honest limits. `reports/` is gitignored, so the annotation artifacts behind §6 and §7 are produced locally rather than shipped with a clone — the labels are annotation data, not a computation. And approximate nearest-neighbour search moves fine-grained aggregates by ≤0.001 between independently built indices; headline A/B/C figures reproduce exactly, and no reported gap or test changes.

Repository: github.com/pouyapd/TrustRAG — paper draft, literature review, reviewer simulation, venue analysis and the full human study are under `docs/paper/`. Comments on the framing in §8, and on whether a single-annotator study reported transparently is publishable, would be especially welcome.